SIMATS ENGINEERING

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ECA1804 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO1 ENGINEERING DESIGN ASSIGNMENT

Standards-Compliant Optical Fiber Link Design: From Communication Requirement to Deployment Decision

Submitted by

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STAGE 1 – INTERPRETING THE COMMUNICATION REQUIREMENT

The service is a 60 km point-to-point link carrying 10 Gb/s traffic from a central office (CO) to a business building, with a later upgrade to 100 Gb/s. Working backward from this requirement, the link must deliver enough optical power, keep pulses separable at the receiver, and keep the signal well above the amplified spontaneous emission (ASE) noise floor, all while surviving tight bends at the building entry.

1.1 The 1550 nm operating window and the nature of light

1550 nm sits in the C-band, where silica fiber has its lowest attenuation (about 0.18–0.2 dB/km) because Rayleigh scattering falls as $1/\lambda^4$ and the window lies between the OH^- absorption peaks and the infrared absorption edge. It is also the gain band of erbium-doped fiber amplifiers (EDFAs), which makes amplification and a later DWDM/100G upgrade straightforward. The penalty is that standard G.652 fiber has a relatively large chromatic dispersion (about 17 ps/(nm·km)) at this wavelength.

Light can be treated as rays only when the core is many wavelengths across (multimode fiber B). For a core of a few micrometres (fiber A) the wave nature dominates: the field must satisfy Maxwell's wave equation and only discrete LP modes propagate. Source coherence (spectral width $\Delta\lambda = 0.4$ nm) directly sets chromatic broadening, and the two orthogonal polarization states of the fundamental mode give rise to birefringence and polarization-mode dispersion (PMD).

1.2 Quantities that govern the design

Area	Quantities to be controlled	Why it matters for this link
Light acceptance / guidance	NA, V-number, LP modes, cut-off wavelength λc	Decides single-mode or multimode operation and thus the bandwidth–distance limit
Fiber category	ITU-T G.652.D, G.655, G.657.A	Sets attenuation, dispersion, bend tolerance and upgrade compatibility
Loss / power budget	Absorption, Rayleigh scattering, bending, connectors, splices	Received power must exceed −20 dBm by at least 6 dB
Pulse integrity	Modal, chromatic and polarization-mode dispersion; DCF compensation	Broadened pulse must stay within the 100 ps bit period (10 ps at 100 Gb/s)
Signal quality	OSNR in 0.1 nm reference bandwidth, Q-factor, BER	OSNR must stay ≥ 17 dB after amplification
Practical constraints	Bend radius at entry, upgrade path, system margin	Fiber choice must remain valid for 100 Gb/s and for installation conditions

1.3 Characteristics of an acceptable solution

- Single-mode propagation at 1550 nm ($V < 2.405$), so modal dispersion is eliminated.
- A standard ITU-T fiber that is widely deployed and upgrade-friendly, with a bend-insensitive section where the cable enters the building.
- Total link loss low enough to leave at least 6 dB of margin above −20 dBm sensitivity.
- Chromatic dispersion reduced so the output pulse fits within the bit period; PMD within 0.1 T_b now and for the upgrade.
- An amplification strategy that restores power without pushing OSNR below 17 dB.

Stage 1 Deliverable – Design Requirement Statement

The proposed link shall carry 10 Gb/s NRZ traffic over 60 km at 1550 nm using a single-mode ITU-T standard fiber with a bend-insensitive building-entry section; it shall deliver received power at least 6 dB above the −20 dBm receiver sensitivity, keep the dispersion-broadened pulse within the 100 ps bit period, keep PMD below 0.1 T_b for both 10 Gb/s and a 100 Gb/s upgrade, and maintain OSNR ≥ 17 dB (0.1 nm reference bandwidth) after any optical amplification.

STAGE 2 – FIBER AND PROPAGATION CONFIGURATION

2.1 Numerical aperture (light-acceptance capability)

$$NA = \sqrt{(n_1^2 - n_2^2)}, \quad \theta a = \sin^{-1}(NA), \quad \Delta = (n_1 - n_2)/n_1$$

Fiber A: NA = $\sqrt{(1.4500^2 - 1.4450^2)} = \sqrt{0.014475} =$ **0.1203**; θa = 6.91°; Δ = 0.345 %.

Fiber B: NA = $\sqrt{(1.480^2 - 1.465^2)} = \sqrt{0.044175} =$ **0.2102**; θa = 12.13°; Δ = 1.01 %.

Fiber B collects about three times more light (NA^2 ratio ≈ 3.05), which matters for LED sources, but a 1550 nm laser launches efficiently into the small NA of fiber A, so light collection is not the limiting factor here.

2.2 V-number (normalized frequency)

$$V = (2\pi a/\lambda) \cdot NA$$

Fiber A: V = (2π × 4.0 / 1.55) × 0.1203 = 16.215 × 0.1203 = **1.95**

Fiber B: V = (2π × 25 / 1.55) × 0.2102 = 101.34 × 0.2102 = **21.30**

2.3 Number of modes and cut-off

Single-mode condition: V < 2.405 (cut-off of the LP_{11} mode). For large V the mode count is approximated by M ≈ $V^2/2$ for step-index and M ≈ $V^2/4$ for graded-index (parabolic) profiles.

Fiber A: V = 1.95 < 2.405, so only the LP_{01} mode propagates – **single-mode**. Cut-off wavelength λc = 2πa·NA/2.405 = (2π × 4.0 × 0.1203)/2.405 = **1.257 µm (1257 nm)**. Since 1550 nm > λc, the fiber is single-mode at the operating wavelength, and λc < 1260 nm is consistent with G.652 practice.

Fiber B: M ≈ $21.30^2/2$ ≈ **227 modes** (step-index); a graded-index fiber of the same V would carry ≈ **113 modes**. Its cut-off wavelength for single-mode operation would be 13.7 µm, so it is strongly multimode at 1550 nm.

2.4 Ray theory versus wave optics

Ray theory is adequate when V is large (V ≫ 10) and the core is many wavelengths wide; fiber B (V = 21.3, 2a = 50 µm ≈ 32λ) can be described by rays bouncing at angles up to θa, and its modal dispersion follows directly from the path-length difference between axial and critical rays. Fiber A (V = 1.95, 2a = 8 µm ≈ 5λ) cannot be described by rays: the field extends into the cladding (mode-field diameter larger than the core) and only the wave-equation solution – the LP_{01} mode – explains guidance, cut-off and dispersion.

2.5 Comparison of propagation alternatives

Intermodal delay over 60 km: step-index Δτ = $Ln_1\Delta/c$ ≈ 50 ns/km × 60 = **3.0 µs**; graded-index Δτ = $Ln_1\Delta^2/(8c)$ ≈ 63 ps/km × 60 = **3.8 ns**. Both are far larger than the 100 ps bit period.

Option	Modal dispersion (60 km)	Bandwidth–distance	Suitability
Single-mode (fiber A)	None (one mode)	Limited only by chromatic dispersion/PMD; >100 Gb/s·km easily	Suitable
Step-index multimode (fiber B)	≈ 3.0 μs	≈ 20 Mb/s·km	Unsuitable (bit rate ≈ 0.2 Mb/s at 60 km)
Graded-index multimode	≈ 3.8 ns	≈ a few Gb/s·km at best	Unsuitable beyond ~1 km at 10 Gb/s

ITU-T category	Key properties at 1550 nm	Role in this link
G.652.D	Standard SMF, low water peak, D ≈ 17 ps/(nm·km), α ≈ 0.18–0.2 dB/km	Main 60 km trunk; most widely installed; fully compatible with coherent 100G
G.655	Non-zero dispersion-shifted, D ≈ 4.5 ps/(nm·km), smaller effective area	Alternative trunk; less CD, but costlier and more nonlinear for DWDM
G.657.A	Bend-insensitive, G.652.D-compatible	Short building-entry section with tight bends

DESIGN DECISION 1 – Fiber / Propagation Configuration

I recommend a **single-mode, step-index fiber of the fiber-A type (V = 1.95, λc = 1257 nm), deployed as ITU-T G.652.D for the 60 km trunk with a G.657.A1 bend-insensitive section at the building entry** propagation/fiber configuration because only single-mode operation removes modal dispersion (fiber B would give ≈ 3 μs spread against a 100 ps bit period), G.652.D is the standard, low-cost, low-water-peak fiber that coherent 100 Gb/s transceivers are designed for, and G.657.A1 is fully splice-compatible with G.652.D while cutting the entry bend loss from 0.8 dB to 0.1 dB.

STAGE 3 – OPTICAL POWER BUDGET

3.1 Total attenuation coefficient

The single coefficient that represents all intrinsic and distributed losses is the fiber attenuation coefficient α (dB/km):

$$\alpha = \alpha_abs + \alpha_scat + \alpha_bend = 0.04 + 0.11 + 0.03 = 0.18\ dB/km$$

$$Fiber\ loss = \alpha \times L = 0.18 \times 60 = 10.8\ dB$$

3.2 Practical link budget

Maximum allowed loss = P_tx − P_sens − margin = 3 − (−20) − 6 = **17 dB**.

Loss element	G.652.D entry (dB)	G.657.A entry (dB)
Fiber, 60 km × 0.18 dB/km	10.80	10.80
Connectors, 4 × 0.4 dB	1.60	1.60
Splices, 10 × 0.08 dB	0.80	0.80
Building-entry bend loss	0.80	0.10
Total link loss	**14.00**	**13.30**
Received power = 3 − loss (dBm)	−11.00	−10.30
Margin over −20 dBm sensitivity (dB)	9.00	9.70
Excess over required 6 dB margin (dB)	+3.00	+3.70

3.3 Dominant loss mechanism

Rayleigh scattering contributes 0.11 × 60 = **6.6 dB**, i.e. 61 % of the fiber loss and 50 % of the total G.657 link loss; absorption gives 2.4 dB and distributed bending 1.8 dB. Rayleigh scattering is intrinsic to glass density fluctuations and cannot be removed after manufacture; it can only be minimized by fiber selection (ultra-low-loss pure-silica-core fiber reaches about 0.15–0.16 dB/km). The largest avoidable losses are the connectors (1.6 dB) and the entry bend (0.8 dB with G.652.D).

3.4 Actions to reduce avoidable loss

- Use G.657.A1 fiber at the building entry and respect minimum bend radius in ducts and trays (saves 0.7 dB).
- Replace mated connectors with fusion splices where re-patching is not required, and use clean, low-loss APC connectors (≤ 0.25 dB) inspected with a fiber microscope; consider fewer, longer cable sections to reduce splice count.

DESIGN DECISION 2 – Power-Budget Configuration

I recommend a **G.652.D trunk with a G.657.A1 building-entry section (total passive loss 13.3 dB, received power −10.3 dBm, 9.7 dB margin)** power-budget configuration

because it meets the 6 dB system margin with 3.7 dB to spare, is 0.7 dB better than an all-G.652.D route, and focuses optimization on the dominant mechanism – Rayleigh scattering (6.6 dB) – through low-loss fiber specification, while connector and bend losses are controlled by installation practice. Note that this margin is before dispersion compensation; Stage 4 shows that the DCF loss must be offset by amplification.

STAGE 4 – DISPERSION AND POLARIZATION MANAGEMENT

4.1 Modal dispersion

In single-mode fiber A only the LP_{01} mode exists, so there is no group-delay difference between modes and modal dispersion is zero. In multimode fiber B, 227 modes travel different path lengths, giving about 50 ns/km of spread. A graded (parabolic) index profile makes the longer, off-axis rays travel through lower-index glass at higher speed, equalizing delays and reducing the spread by a factor of about $\Delta/8$ (to ≈ 63 ps/km), which is still too large for 60 km.

4.2 Chromatic-dispersion broadening

$$\Delta\tau = D \cdot \Delta\lambda \cdot L, \qquad \tau_out = \sqrt{(\tau_in^2 + \Delta\tau^2)}$$

Fiber	$\Delta\tau = D \cdot \Delta\lambda \cdot L$	τ_out (RSS with 30 ps)	τ_out / T_b	Assessment
G.652.D (17 ps/(nm·km))	17 × 0.4 × 60 = 408 ps	$\sqrt{(30^2 + 408^2)}$ = 409.1 ps	4.09	Severe ISI – pulse spreads over ≈ 4 bit slots
G.655 (4.5 ps/(nm·km))	4.5 × 0.4 × 60 = 108 ps	$\sqrt{(30^2 + 108^2)}$ = 112.1 ps	1.12	Marginal – exceeds T_b; eye partly closed

With G.652.D, each pulse overlaps three to four neighbouring bit slots, so the eye would be closed and BER unacceptable. G.655 is close to the limit but still exceeds 100 ps, and a practical design usually requires the broadened pulse to stay below about 0.7 T_b; therefore neither fiber works uncompensated at 60 km with this 0.4 nm source.

4.3 Dispersion compensation

Complete first-order compensation requires the accumulated dispersion to sum to zero:

$$D_SMF \cdot L_SMF + D_DCF \cdot L_DCF = 0$$

$$L_DCF = (17 \times 60) / 100 = 10.2 \text{ km}$$

DCF loss = 10.2 × 0.5 = **5.1 dB**. New total loss (G.657 entry) = 13.3 + 5.1 = **18.4 dB**; received power = 3 − 18.4 = **−15.4 dBm**; margin over sensitivity = **4.6 dB**, which is **1.4 dB short** of the required 6 dB. (With a G.652.D entry: 19.1 dB, −16.1 dBm, 3.9 dB margin.) Hence DCF on its own breaks the power budget and must be combined with an optical amplifier.

For comparison, a G.655 trunk needs only 4.5 × 60 / 100 = 2.7 km of DCF (1.35 dB), leaving −11.65 dBm and 8.35 dB margin without amplification. This is attractive for 10 Gb/s, but G.655 is more expensive, less available, and offers no advantage for a coherent 100 Gb/s upgrade, where dispersion is compensated electronically in the DSP.

4.4 Polarization-mode dispersion

$$PMD_link = 0.1\ ps/\sqrt{km} \times \sqrt{60\ km} = 0.775\ ps$$

Bit rate	T_b	PMD limit (0.1 T_b)	Link PMD	Result
10 Gb/s	100 ps	10 ps	0.775 ps (0.84 ps incl. 10.2 km DCF)	Acceptable – 7.7 % of limit
100 Gb/s upgrade	≈ 10 ps	1 ps	0.775 ps (0.84 ps incl. DCF)	Acceptable but tight – 78–84 % of limit

Mitigation: specify low-PMD fiber (link design value well below 0.1 ps/√km) and, for the 100 Gb/s upgrade, use coherent dual-polarization transceivers whose DSP adaptively equalizes PMD; avoiding cable stress and vibration during installation also keeps PMD low.

4.5 Birefringence and beat length

$$L_B = \lambda / B$$

Standard fiber: L_B = $1.55 \times 10^{-6} / 1 \times 10^{-7}$ = **15.5 m**. **PM fiber:** L_B = $1.55 \times 10^{-6} / 4 \times 10^{-4}$ = **3.875 mm**.

The very short beat length of PM fiber means its built-in birefringence overwhelms random perturbations, so a launched linear polarization is preserved. This is needed for polarization-sensitive devices (modulator inputs, interferometric sensors, coherent local-oscillator pigtails), but not for this transmission link: the receiver is polarization-insensitive at 10 Gb/s and the coherent 100G receiver tracks polarization digitally. PM fiber is also costlier and lossier, so it is **not** used in the 60 km span.

DESIGN DECISION 3 – Dispersion / PMD-Management Configuration

I recommend a **G.652.D trunk with 10.2 km of DCF (D = −100 ps/(nm·km)) for full first-order compensation at 10 Gb/s, standard (non-PM) low-PMD single-mode fiber, and electronic CD/PMD compensation in coherent DSP for the 100 Gb/s upgrade** dispersion/PMD configuration because the uncompensated pulse would grow to 409 ps (4.1 T_b), the 10.2 km DCF brings net dispersion to zero at a cost of 5.1 dB (recovered by the amplifier in Stage 5), PMD is only 0.775 ps against limits of 10 ps (10 Gb/s) and 1 ps (100 Gb/s), and the G.652.D choice keeps the plant ready for coherent 100 Gb/s, where the DCF can simply be bypassed.

STAGE 5 – OSNR AND AMPLIFICATION STRATEGY

5.1 Performance metric

Received power alone does not show signal quality, because an amplifier boosts noise as well. The correct metric is the **optical signal-to-noise ratio (OSNR)** – the ratio of signal power to ASE noise power measured in a 0.1 nm reference bandwidth (≈ 12.5 GHz at 1550 nm), measured with an optical spectrum analyser (interpolating the noise floor either side of the channel, or by in-band/polarization-nulling methods).

5.2 OSNR before and after amplification

$$OSNR\ (dB) = P_signal\ (dBm) - P_ASE\ (dBm)$$

Quantity	Before amplifier	After amplifier
Signal power	−16 dBm	−16 + 14 = −2 dBm
ASE noise power (0.1 nm)	−39 dBm	−39 + 16 = −23 dBm
OSNR	−16 − (−39) = **23 dB**	−2 − (−23) = **21 dB**
Required OSNR	17 dB	17 dB
Remaining margin	6 dB	**4 dB**

The amplifier raises signal power by 14 dB but degrades OSNR by 2 dB, because the ASE noise grows 2 dB more than the signal. The resulting 21 dB still exceeds the 17 dB requirement with 4 dB of margin. The −16 dBm study point is also consistent with the −15.4 dBm signal expected after the DCF, so the OSNR study represents the actual amplifier position in the proposed link.

5.3 Effect of inadequate OSNR

If OSNR fell below 17 dB, signal–ASE beat noise would broaden the "1" and "0" levels, lowering the Q-factor. Since BER ≈ ½·erfc(Q/√2), a small Q drop causes a large BER rise: Q = 6 gives BER ≈ 10^{-9}, while Q = 4.75 gives only about 10^{-6}. The system margin would disappear even if received power looked adequate, and no amount of extra gain would recover it.

5.4 Measures to maintain or improve OSNR

- Use low-noise-figure EDFAs (NF ≈ 4–5 dB) and place amplification where the signal is still strong, e.g. combine a booster at the transmitter with a mid-stage DCF so the noise is added at a higher signal level.
- Shorten amplifier spacing or add distributed Raman gain, and optimize launch power below the nonlinear threshold; use narrow optical filters after the amplifier to remove out-of-band ASE.

DESIGN DECISION 4 – OSNR / Amplification Strategy

I recommend **accepting a single 14 dB EDFA used as a receiver-end pre-amplifier after the DCF, with a variable optical attenuator (VOA) to set the received power inside the receiver window** amplification/OSNR strategy because it recovers the 5.1 dB DCF loss (received power rises from −15.4 dBm to about −1.4 dBm, far above −20 dBm), while OSNR only drops from 23 dB to 21 dB, leaving 4 dB above the required 17 dB – enough to sustain BER $\leq 10^{-9}$ at 10 Gb/s.

STAGE 6 – VERIFICATION OF THE COMPLETE LINK

Requirement	Specified / Target	Achieved	Satisfactory?
Modal condition / fiber category	Single-mode at 1550 nm; ITU-T fiber	$V = 1.95 < 2.405$, $\lambda c = 1257$ nm; G.652.D + G.657.A1	Yes
Fiber loss / link loss	≤ 17 dB (3 + 20 − 6)	$\alpha = 0.18$ dB/km; 10.8 dB fiber; 13.3 dB passive; 18.4 dB incl. DCF; 4.4 dB net after 14 dB gain	Yes (with EDFA)
Received optical power and margin	≥ −14 dBm (6 dB above −20 dBm)	≈ −1.4 dBm before VOA; 18.6 dB above sensitivity	Yes
Pulse broadening / DCF requirement	$\tau_out \leq T_b = 100$ ps	409 ps uncompensated → ≈ 30 ps (net D = 0) with 10.2 km DCF	Yes
PMD / polarization performance	≤ 10 ps (10G); ≤ 1 ps (100G)	0.775 ps (0.84 ps incl. DCF); PM fiber not needed	Yes (tight at 100G)
OSNR	≥ 17 dB (0.1 nm)	21 dB after amplification (4 dB margin)	Yes

All values are consistent with the Stage 1 Design Requirement Statement. The only requirement that is not met by the passive design alone is the 6 dB margin once the DCF is inserted (4.6 dB); this is exactly why the amplifier is part of the configuration rather than an optional extra. With the residual dispersion after DCF close to zero, the output pulse returns to roughly the 30 ps input width, so ISI is negligible at 10 Gb/s.

FINAL SYSTEM DESIGN (K6 – CREATE / DESIGN)

Integrated link configuration

The design combines the four decisions into one link. A 1550 nm, +3 dBm, 10 Gb/s transmitter in the central office feeds the CO patch panel and a 60 km G.652.D trunk with ten fusion splices. At the building, a G.657.A1 bend-insensitive section carries the fiber through the tight entry path. In the building equipment room, a 10.2 km DCF module fully compensates the trunk dispersion, followed by a 14 dB EDFA pre-amplifier and a VOA that sets the power at the 10 Gb/s receiver. Each choice supports the others: single-mode G.652.D removes modal dispersion, the DCF removes chromatic dispersion, the EDFA pays back the DCF loss, and the OSNR check confirms that the amplifier does not trade power for noise.

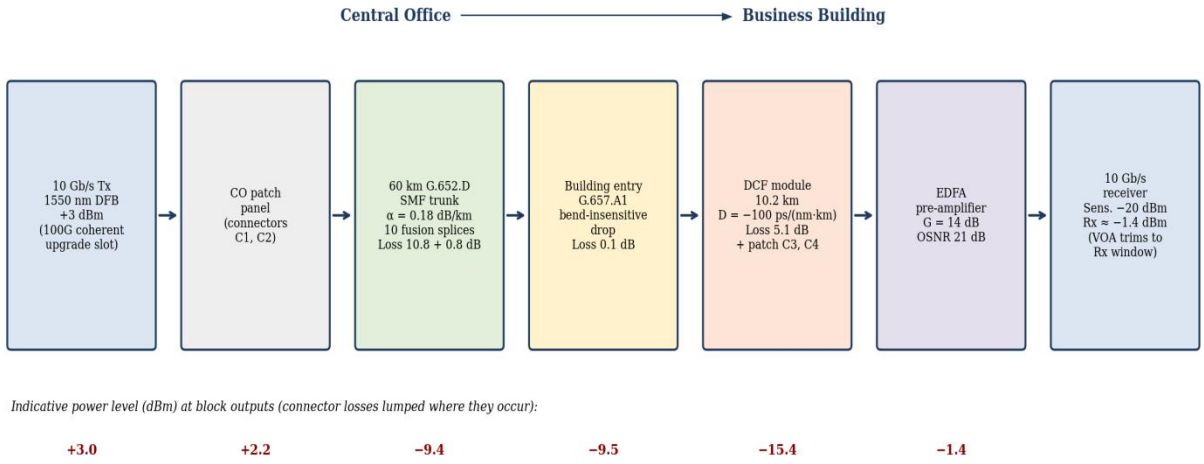


Figure 1 – Block diagram of the proposed 60 km, 1550 nm optical fiber link

Final parameter / design table

Design element	Adopted value / choice
Propagation / fiber	Single-mode step-index (V = 1.95, NA = 0.120, a = 4 μm, λc = 1257 nm)
ITU-T category	G.652.D trunk (60 km); G.657.A1 at building entry
Attenuation coefficient	0.18 dB/km (Rayleigh 0.11 dominant)
Passive link loss	13.3 dB (fiber 10.8, connectors 1.6, splices 0.8, entry bend 0.1)
Dispersion compensation	DCF, 10.2 km, D = −100 ps/(nm·km), loss 5.1 dB; net dispersion ≈ 0 ps/nm
PMD management	Low-PMD fiber; link PMD 0.775 ps; coherent DSP for 100G; no PM fiber
Amplifier setting	EDFA pre-amplifier, gain 14 dB, after DCF; input ≈ −15.4 dBm
OSNR	23 dB before, 21 dB after amplifier (required 17 dB)
Received power	≈ −1.4 dBm, trimmed with VOA to the receiver dynamic range
Upgrade path	100 Gb/s coherent DP-QPSK on the same G.652.D; DCF bypassed, CD/PMD handled in DSP

FINAL ENGINEERING DECISION – WOULD I DEPLOY THIS DESIGN?

YES – the integrated design (G.652.D + G.657.A1 + 10.2 km DCF + 14 dB EDFA pre-amplifier) should be deployed.

- **Fiber/modal suitability:** V = 1.95 guarantees single-mode operation at 1550 nm, and G.652.D/G.657.A1 are standard, mutually compatible categories.
- **Propagation and bandwidth–distance:** modal dispersion is eliminated; multimode fiber B (≈ 3 μs spread) was correctly rejected.

- **Loss and received power:** passive loss of 13.3 dB gives −10.3 dBm; with the DCF and amplifier the receiver sees about −1.4 dBm, far above −20 dBm plus the 6 dB margin.
- **Dominant loss and optimization:** Rayleigh scattering (6.6 dB) dominates and is addressed through low-loss fiber specification; connector and bend losses are kept low by splicing and G.657.A1.
- **Dispersion and PMD:** 408 ps of chromatic broadening is fully compensated by 10.2 km of DCF; PMD (0.775 ps) is within both the 10 Gb/s and 100 Gb/s limits.
- **OSNR:** 21 dB after amplification leaves a 4 dB margin over 17 dB.
- **Overall:** the link meets every 10 Gb/s requirement and keeps a clean upgrade path to coherent 100 Gb/s.

The decision would be NO only for the passive-only version, since inserting the DCF without an amplifier leaves 4.6 dB margin, 1.4 dB short of the 6 dB requirement. Two conditions are attached to deployment: (1) the receiver overload level must be checked and the VOA set so the received power sits inside the receiver window (e.g. about −8 dBm), and (2) PMD of the installed cable should be measured on acceptance, because at 100 Gb/s the margin to the 1 ps limit is small. If the operator later prefers to avoid an amplifier at 10 Gb/s, the alternative redesign would be a G.655 trunk with 2.7 km of DCF (8.35 dB margin, no EDFA), at the cost of a less common fiber type for the upgrade.

ENGINEERING DESIGN PROCESS SUMMARY

Requirement (60 km, 10 → 100 Gb/s, 1550 nm) → Alternatives (fiber A vs B; G.652.D vs G.655 vs G.657) → Analysis (NA, V, modes, loss, dispersion, PMD, OSNR) → Selection (single-mode G.652.D + G.657.A1) → Link design (DCF 10.2 km + 14 dB EDFA) → Verification (all six criteria satisfied) → Deployment decision (YES, with VOA setting and PMD acceptance test).